# Dr Michael B. Hall

Genomic Microbiologist · Bioinformatician

Sunshine Coast, Australia · mhaldini@icloud.com · mbhall88.github.io · ORCID 0000-0003-3683-6208 · GitHub @mbhall88 · Scholar · LinkedIn

Bioinformatician and genomic microbiologist developing open-source software and long-read sequencing methods for the diagnosis and surveillance of drug-resistant bacterial infections. First-author work spans tuberculosis diagnostics, antimicrobial-resistance genomics, and the benchmarking of long-read variant calling.

## APPOINTMENTS

2025–present **Postdoctoral Research Fellow (Level B.2), Bacterial Genomics Group (Dr Leah Roberts)** — The University of Queensland, Frazer Institute, Brisbane, AU

– Lead bioinformatician on a clinical antimicrobial-resistance genomics program.

2022–2025 **Postdoctoral Research Fellow (Level B.1), Coin Group** — Peter Doherty Institute for Infection and Immunity, The University of Melbourne, Melbourne, AU

– Led a multi-institutional bacterial Nanopore variant-calling benchmark and developed resource-efficient Kraken databases for host-read removal and mycobacterial classification from clinical metagenomes.

2021–2022 **Senior Scientific Officer** — Child Health Research Centre, The University of Queensland, Brisbane, AU

– Analysed Nanopore direct-RNA sequencing data to distinguish bacterial from viral sepsis via RNA-modification signatures; benchmarked deep-learning models for motif-specific signal perturbation.

2017–2021 **PhD Candidate (EMBL International PhD Programme)** — University of Cambridge / EMBL-EBI, Cambridge, UK

– Doctoral research on bacterial genomics with genome graphs and long-read sequencing; trained 25+ workshop participants in UNIX, Bash, and containers.

## EDUCATION

2017–2022 **PhD**, Bioinformatics — University of Cambridge

– EMBL International PhD Programme — a competitive four-year international doctoral fellowship at the European Bioinformatics Institute (EMBL-EBI); supervisor Zamin Iqbal.

2016–2017 **Master of Bioinformatics (Research Extensive)**, Bioinformatics — The University of Queensland

– Graduated with a 7.0 GPA; awarded the Master of Molecular Biology and Master of Bioinformatics Prize for the highest cumulative GPA in either program.

2013–2015 **Bachelor of Biomedical Science**, Biomedical Science — Queensland University of Technology

– Graduated with Distinction; 7.0 GPA.

## PUBLICATIONS

2026 **Hall MB**, Xue Y, Lee TSE, Herring E, Hume J, Wick RR, Kidd T, Runnegar N, Harris PNA, Graves B, Roberts LW. **Novel transposon Tn8026 contributes to the global spread of transmissible linezolid resistance in Enterococcus via a linear plasmid**. *Microbial Genomics* (2026) · first author, corresponding · doi

– Independently taken up internationally: the Public Health Agency of Canada's nosocomial surveillance program adopted the Tn8026 reference and the linear-plasmid assembly method.

2026 Furqon ADC, Roberts LW, **Hall MB**. **Efficient downsampling of genome alignments with Rasusa**. *GigaByte* (2026) · senior author, corresponding · doi

– First senior/last-authored paper; a master's student first-authored it from summer project work under my primary supervision.

2024 **Hall MB**, Wick RR, Judd LM, Nguyen ANT, Steinig EJ, Xie O, Davies MR, Seemann T, Stinear TP, Coin LJM. **Benchmarking reveals superiority of deep learning variant callers on bacterial nanopore sequence data**. *eLife* (2024) · first author, corresponding · doi

– Led a multi-institutional bacterial Nanopore benchmarking study as first and corresponding author; found that current deep-learning callers outperformed Illumina-based calling.

2024 **Hall MB**, Coin LJM. **Pangenome databases improve host removal and mycobacteria classification from clinical metagenomic data**. *GigaScience* (2024) · first author, corresponding · doi

– Built resource-efficient custom Kraken databases optimised for host-read removal and mycobacterial classification from clinical metagenomes; released as NoHuman.

2023 **Hall MB**, Rabodoarivelo MS, Koch A, Dippenaar A, George S, Grobbelaar M, Warren R, Walker TM, Cox H, Gagneux S, Crook D, Peto T, Rakotosamimanana N, Grandjean Lapierre S, Iqbal Z. **Evaluation of Nanopore sequencing for Mycobacterium tuberculosis drug susceptibility testing and outbreak investigation: a genomic analysis**. *The Lancet Microbe* (2023) · first author · doi

– Adopted into the Madagascar National TB Programme's routine diagnostic workflow; first validation that Nanopore sequencing gives clinically actionable TB drug-susceptibility results.

2023 Nilgiriwala K, Rabodoarivelo MS, **Hall MB**, Patel G, Mandal A, Mishra S, et al.. **Genomic Sequencing from Sputum for Tuberculosis Disease Diagnosis, Lineage Determination, and Drug Susceptibility Prediction**. *Journal of Clinical Microbiology* (2023) · joint-first author · doi

– Cited in the WHO TB sequencing guidelines (3rd ed.) as evidence that current sputum sequencing protocols are not yet sufficient for clinical use.

2023 **Hall MB**, Lima L, Coin LJM, Iqbal Z. **Drug resistance prediction for Mycobacterium tuberculosis with reference graphs**. *Microbial Genomics* (2023) · first author, corresponding · doi

– Reference-graph approach to TB drug-resistance prediction; released as DrPRG. Won ABACBS Best Lightning Talk (2022).

2022 **Hall MB**, Coin LJM. **Assessment of the 2021 WHO Mycobacterium tuberculosis drug resistance mutation catalogue on an independent dataset**. *The Lancet Microbe* (2022) · first author, corresponding · doi

– Directly influenced the 2023 WHO TB mutation catalogue to caution against overfitting and emphasise independent validation.

2022 Ganesamoorthy D, Robertson AJ, Chen W, **Hall MB**, Cao MD, Ferguson K, Lakhani SR, Nones K, Simpson PT, Coin LJM. **Whole genome deep sequencing analysis of cell-free DNA in samples with low tumour content**. *BMC Cancer* (2022) · doi

– Performed fragment-size analysis of deep whole-genome sequencing data from plasma cell-free DNA in a breast cancer study.

2022 **Hall MB**. **Rasusa: Randomly subsample sequencing reads to a specified coverage**. *Journal of Open Source Software* (2022) · first author, corresponding · doi

– Sole-author paper for Rasusa (>77k installs); integrated into the Bactopia and Dragonflye pipelines.

2021 Colquhoun RM, **Hall MB**, Lima L, Roberts LW, Malone KM, Hunt M, Letcher B, Hawkey J, George S, Pankhurst L, Iqbal Z. **Pandora: nucleotide-resolution bacterial pan-genomics with reference graphs**. *Genome Biology* (2021) · doi

– Co-led Pandora's development over three years; led the variant-calling engine, novel-variant logic, and the benchmarking and validation.

2021 Urban L, Holzer A, Baronas JJ, **Hall MB**, …, Stammnitz MR. **Freshwater monitoring by nanopore sequencing**. *eLife* (2021) · doi

– Conducted all metagenomic sequencing data analysis; field-weighted citation impact 5.09 (top 3% worldwide).

2021 Mölder F, Jablonski KP, Letcher B, **Hall MB**, …, Köster J. **Sustainable data analysis with Snakemake**. *F1000Research* (2021) · doi

– Official maintainer of Snakemake; added job-scheduler support (Slurm, LSF), Bash/Rust compatibility, and the snakefmt formatter. Field-weighted citation impact 72.84.

## SOFTWARE

**Snakemake** — Workflow management system for reproducible data analysis. (Python, 1,885,581 installs)

– Co-author and official maintainer; added Slurm/LSF scheduler support and native Bash/Rust scripts to a workflow system installed >1.8M times.

**snakefmt** — Code formatter for Snakemake workflows. (Python, 115,808 installs)

– Formatter for the Snakemake ecosystem; >115,000 installs.

**Rasusa** — Randomly subsample sequencing reads to a target coverage or read count. (Rust, 77,123 installs)

– >77,000 installs; a top-five bioinformatics tool written in Rust.

**TBpore** — Nanopore-based Mycobacterium tuberculosis detection and resistance prediction. (Python, 23,372 installs)

– Adopted into the Madagascar National TB Programme's routine diagnostic workflow (~15 TB samples/yr; ~200 patients to date).

**NoHuman** — Remove human reads from sequencing data using a pangenome graph. (Rust, 12,020 installs)

– Pangenome-based host-read removal for clinical metagenomics; integrated into Bactopia.

**LRGE** — Long-read genome-size estimation without assembly. (Rust, 11,288 installs)

– Embedded in the Hybracter and Autocycler assemblers; won ABACBS Best Lightning Talk (2025).

**DrPRG** — Drug-resistance prediction for Mycobacterium tuberculosis via genome graphs. (Rust, 5,599 installs)

– Reference-graph TB drug-resistance prediction.

**Pandora** — Nucleotide-resolution bacterial pan-genomics with reference graphs. (C++, 4,884 installs)

– Co-led development over three years; led the variant-calling engine and novel-variant logic.

## GRANTS & FUNDING

2024 — **AllTheBacteria dataset hosting (AWS Open Data Sponsorship Program)** — Amazon Web Services (Project lead, active)

– Applied for, secured and coordinated indefinite AWS Open Data hosting for 9 TB of AllTheBacteria data: 2.4 million bacterial genome assemblies and a LexicMap index for rapid sequence search.

2022 — **AIDS and Infectious Disease Network grant (Madagascar TB clinical trial)** — Fonds de Recherche du Québec en Santé (FRQS) (Contributor, completed)

– Co-developed laptop-compatible computational workflows for a TB clinical trial in a remote region of Madagascar.

2017–2021 — **EMBL International PhD Programme fellowship** — European Molecular Biology Laboratory (EMBL) (Recipient, completed)

– Awarded a competitive four-year international PhD fellowship by EMBL.

## AWARDS & RECOGNITION

2025 — **ABACBS Best Lightning Talk** — Australian Bioinformatics & Computational Biology Society (national)

2024 — **MicroSeq Best Lightning Talk** — MicroSeq (national)

2022 — **ABACBS Best Lightning Talk** — Australian Bioinformatics & Computational Biology Society (national)

2018 — **Master of Molecular Biology and Master of Bioinformatics Prize** — The University of Queensland (institutional)

– Awarded for the highest cumulative GPA in the Master of Molecular Biology or Master of Bioinformatics programs; $500 prize.

2016–2017 — **Dean's Commendation for Academic Excellence (three semesters)** — The University of Queensland (institutional)

2016 — **Westpac Future Leaders Scholarship** — Westpac Bicentennial Foundation (national)

– Inaugural 2016 scholar; the scholarship supported focused master's study, leadership development, new professional networks and international study, followed by UQ's prize for the highest cumulative GPA in the Master of Molecular Biology or Master of Bioinformatics programs.

2015 — **Faculty of Health Equity Award** — Queensland University of Technology (institutional)

2015 **Executive Dean's Commendation (Semester 2)** — Queensland University of Technology (institutional)

2014 **Wolters Kluwer Prize for Outstanding Student Achievement** — Queensland University of Technology (institutional)

2013 **Dean's Commendation (Semester 1)** — Queensland University of Technology (institutional)

2026 **Publication of the Month Award, Early Career Researcher category (July 2026)** — Frazer Institute, The University of Queensland (institutional)

– Awarded for the first-authored Microbial Genomics paper on Tn8026 and transmissible linezolid resistance, recognising its publication quality, impact and research outcomes.

## INVITED TALKS

2026 **Long-read plasmid assembly (invited workshop co-lead)** — Australian Society for Microbiology Annual Scientific Meeting — Plasmid Workshop, Australia · workshop

– Invited co-lead of the plasmid-assembly component of a full-day national workshop (134 registrants), off the back of the Tn8026 linear-plasmid preprint.

2026 **Straight plasmids in the land of banana benders (and the potential futility of adapter trimming)** — Oxford Nanopore Technologies User Group Meeting, Brisbane, AU · invited

– Invited talk hosted by the sequencing-platform provider whose technology underpins the work.

2025 **LRGE: long-read genome-size estimation** — ABACBS Annual Conference, Australia · contributed

– Lightning talk; won ABACBS Best Lightning Talk.

2024 **Benchmarking deep-learning variant callers on bacterial nanopore data** — Australian Society for Microbiology Annual Scientific Meeting, Australia · contributed

2024 **Bacterial nanopore variant calling** — ASM Hour seminar series, Online (Australia) · invited

2023 **DrPRG: TB drug-resistance prediction with reference graphs** — International Congress of Genetics, Australia · contributed

2023 **DrPRG and TB diagnostics** — NSW Health TB conference, Australia · contributed

2023 **TB genomics and diagnostics** — Queensland Mycobacterial Reference Laboratory, QLD, AU · invited

2022 **Reference-graph methods for bacterial genomics** — Australasian Genomic Technologies Association, Australia · contributed

2019 **Pandora: bacterial pan-genomics with reference graphs** — Computational Pangenomics Workshop, University of Bielefeld, Bielefeld, Germany · invited

– Invited international talk on Pandora.

2026 **Tn8026 and transmissible linezolid resistance in Enterococcus** — ABACBS Virtual EMCR Symposium 2026, Online (Australia) · contributed

## SUPERVISION & MENTORING

2026 **Undergraduate (Primary)**, The University of Queensland

– Project: 'Assembling small plasmids from Nanopore read length distributions'.

– Completed primary supervision of an undergraduate research project on plasmid assembly.

2025–2026 **Master's (Primary)** — Dimas, The University of Queensland

– First-authored 'Efficient downsampling of genome alignments with Rasusa' (GigaByte 2026) from summer project work.

– Primary supervisor; the student first-authored a peer-reviewed paper under supervision — a strong mentoring and emerging-senior-authorship anchor.

2025–2026 **Master's (Primary)** — Wenjie Song, UQ (UQCCR / Frazer Institute)

– Thesis: 'The Escape of Human Genomic Data into Public Repositories'.

– First completed primary master's supervision.

2026–present **Master's (Primary)**, The University of Queensland

– Thesis: 'Evaluating the impact of quality-control preprocessing on Oxford Nanopore sequencing data analysis'.

– Primary supervisor on a current master's project evaluating ONT quality-control preprocessing.

2024–2026 **Master's (Co-primary)**, University of Melbourne

– Completed master's degree, including the thesis 'Detection of novel antimicrobial resistance-conferring mutations in Mycobacterium tuberculosis', in July 2026.

– Co-primary supervisor of a completed master's project investigating previously unrecognised causes of antibiotic resistance in tuberculosis.

2025–present **MPhil (Secondary)**, The University of Queensland

– Investigating plasmid evolution and transmission — long-term dynamics of OXA-181 resistance plasmids in Brisbane to inform genomic surveillance.

2023–present **PhD (Advisory)**

– PhD advisory committee member.

**Postdoctoral (Mentor)**, Doherty Institute

– Mentored a postdoctoral fellow in computational-analysis pipeline development.

2024–present **Scholar (Mentor)**, Westpac Scholars program

– Monthly mentor to a scholar through the national Westpac Scholars program.

2017–2021 **PhD (Mentor)**, EMBL-EBI, Wellcome Genome Campus (Hinxton)

– 25+ trainees (10 EMBL + 15 Wellcome) in UNIX, Bash, and containers.

– Trained 25+ incoming PhD students in computational skills during doctoral studies.

## SERVICE & LEADERSHIP

2019–present Reviewer for 9 international journals (12+ manuscripts), including Microbial Genomics, Bioinformatics, Emerging Infectious Diseases and PLOS Global Public Health.

2024–present **Bioconda** — Member, Bioconda Rust team — authored the official Rust packaging guidelines and embedded Rust into Bioconda's core build/lint tooling.

2019–present **Bioconda** — Contributor and recipe maintainer, bioconda-recipes; reviewed/contributed to 190+ community recipe submissions.

2026–present **UQ Frazer Institute** — Member and sponsorship lead, Frazer Institute ECR Committee — leading sponsor outreach for the inaugural Frazer Institute ECR symposium; organising speakers and the 'Research Impact: Planning and Tracking' session for the ECR Professional Development Day in October 2026.

2026–present **UQ Frazer Institute** — Writer of the highlight section of the Frazer Institute ECR newsletter — interviewing a colleague each issue so the institute knows who its researchers are; one profile, reposted by the communications team, led to an external researcher making contact and a new collaboration for that colleague.

2026–present **UQ Frazer Institute** — Co-founder and co-lead, Frazer Institute ECR Skill Swap series — a fortnightly, transferable-skills alternative to the standard ECR seminar series that typically attracts 40–50 attendees.

2025–present **UQ Frazer Institute** — Co-lead, UQCCR / Frazer Institute ONT Working Group — a newly-formed group on Oxford Nanopore applications.

2025 **UQ Frazer Institute** — Organising committee (panel-session lead), UQCCR Infectious Disease Showcase — assembled and coordinated the panel discussion.

2022–2025 **University of Melbourne** — HPC champion (1 of 25 university-wide) — supported high-performance-computing adoption across Doherty researchers; restructured a ~2,000-sample analysis from a projected two months to 1.5 hours.

2023 **Australian Bioinformatics & Computational Biology Society** — Organising committee, ABACBS 2023 annual conference (~400 attendees) — coordinated workshops.

2023 Coordinator, Clinical Informatics Symposium (~50 attendees, 6 workshops).

**ABACBS** — Member, Australian Bioinformatics & Computational Biology Society.

**ASM** — Member, Australian Society for Microbiology.

## RESEARCH IMPACT

2026 **Public Health Agency of Canada — CNISP VRE working group**: Canada's national nosocomial-surveillance programme used the Tn8026 characterisation to interpret linezolid resistance and adopted the linear-plasmid assembly method developed for the study as its recommended protocol.

2026 **Queensland Infection Prevention and Control Unit / Pathology Queensland / Princess Alexandra Hospital**: In collaboration with Pathology Queensland and Princess Alexandra Hospital, the linezolid-resistant Enterococcus investigation contributed to the QIPCU mandate for whole-genome sequencing of every Queensland LRE isolate and supported single-room isolation of colonised patients at Princess Alexandra Hospital.

2023 **Madagascar National TB Programme (NTCPM) / Institut Pasteur Madagascar**: The Madagascar National TB Programme adopted Nanopore sequencing with TBpore into its routine diagnostic workflow, changing clinical management (resolving phenotype/genotype resistance discordance; confirming pre-XDR-TB) and extending to other pathogens.

2024 **World Health Organization**: Cited in the WHO TB sequencing guidelines (3rd ed.) as evidence that current sputum sequencing protocols are not yet sufficient for clinical use (two WHO policy citations).

2023 **World Health Organization**: Influenced the 2023 WHO M. tuberculosis mutation catalogue (2nd ed.) to caution against overfitting and emphasise independent validation.

**Bactopia; Dragonflye**: Rasusa integrated into the Bactopia and Dragonflye bacterial-genomics pipelines for read preparation.

2026 **Broad Institute — viral-ngs**: The Broad Institute adopted Rasusa v3's alignment-based downsampling in its viral-ngs toolkit after testing showed 13-fold lower peak memory use and 17-fold faster analysis with little loss of assembly quality.

2024 **Hybracter; Autocycler**: LRGE embedded in the Hybracter and Autocycler long-read genome assemblers for genome-size estimation.

**Bactopia**: NoHuman and Rasusa integrated into the Bactopia pipeline for automated host-read removal and read preparation, respectively.

**Bioconda community**: Authored the Bioconda Rust packaging guidelines now used as the community standard for distributing Rust-based bioinformatics tools (>240 tools, >4.4M combined downloads).

2026 **Queensland Infection Prevention and Control Unit (Queensland Health)**: Analysis of the linezolid-resistant Enterococcus outbreak (mechanism of resistance acquisition + transmission-chain reconstruction across Queensland) contributed to the QIPCU decision to mandate whole-genome sequencing of all LRE isolates in Queensland.

2026 **Princess Alexandra Hospital — Infection Management Services**: Genomic findings (imported vs locally transmitted isolates; poxtA on a mobile linear plasmid) directly informed enhanced infection-control measures at Princess Alexandra Hospital — including single-room isolation of colonised patients, a step not justifiable on phenotype alone.

Citation and download metrics as of 2026-08.